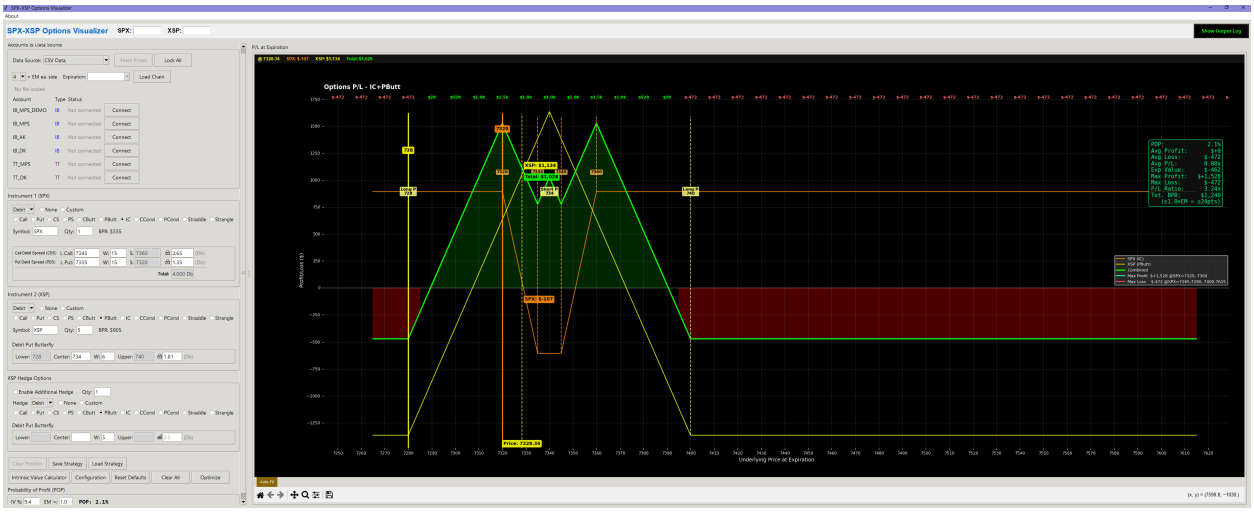
tradebrary

SPX-XSP Options Visualizer

User Guide

Full Reference Manual



Application	SPX-XSP Options Visualizer
Document type	Full User Guide
Instruments	SPX / XSP
Platform	Windows 10 / 11

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Table of Contents

Table of Contents 2

1 Overview 5

1.1 Supported Strategies . . . 5

1.2 Instruments . . . 6

2 Installation 7

2.1 System Requirements . . . 7

2.2 Installing from the Installer . . . 7

2.3 Licensing . . . 7

3 Interface Overview 8

3.1 Left Control Panel (top to bottom) . . . 8

3.2 Chart Area . . . 8

3.3 Menu Bar . . . 9

4 Data Sources 10

4.1 CSV Data . . . 10

4.2 Data From Brokers . . . 11

4.3 Saving Chain Files . . . 11

5 Fetching Prices & Streaming 11

5.1 Fetch Prices Button . . . 12

5.2 Live Streaming . . . 12

5.3 Price Locks . . . 12

6 SPX Strategy Setup 13

6.1 Selecting a Strategy . . . 13

6.2 Strike Entry . . . 13

6.3 Custom Strategy (4 Legs) . . . 14

6.4 BPR Display . . . 14

7 XSP Strategy Setup 15

8 Additional XSP Hedge 16

8.1 Enabling the Hedge . . . 16

8.2 Hedge in the Chart . . . 16

9 Chart Interaction 17

9.1 Reading the Chart . . . 17

9.2 Dragging Strikes . . . 17

9.3 POP and Statistics Box . . . 17

10 POP & IV Settings 18

11 Strategy Optimizer 19

11.1 Optimizer Parameters Window . . . 19

11.2 Running the Optimizer . . . 20

11.3 Optimization Targets . . . 20

11.4 EM Bounds Filtering . . . 20

11.5 Saving and Loading Parameters . . . 20

11.6 Optimizer Results Files . . . 20

12 Saving & Loading Strategies 21

12.1 Auto-Save . . . 21

12.2 Named Strategies . . . 21

13 Configuration 22

13.1 Data Directory . . . 22

13.2 Default Strikes . . . 23

13.3 Chart Settings . . . 23

13.4 Commissions and Slippage . . . 23

13.5 Adding Broker Accounts . . . 24

13.6 Save Config . . . 27

14 Examples 28

14.1 Mountain Range 28
14.2 Hedged Put Spread 29
14.3 Mutant Butterfly 31

15 Quick Reference **32**

15.1 Typical Workflow 32
15.2 Optimizer Quick-Start 32
15.3 Drag-Line Behaviour Summary 32
15.4 Common Issues 33

Appendix A File & Folder Reference **34**

Appendix B Strategy Code Reference **34**

1 Overview

The SPX-XSP Options Visualizer is a desktop application for analysing multi-leg options strategies on two correlated instruments: **SPX** (S&P 500 index options) and **XSP** (mini-SPX options, 1/10th the size of SPX).

The core idea is to view an **SPX** options position hedged with **XSP** options in order to shape a risk graph you are more comfortable with — one that may let you hold the position longer, and with different risk/reward parameters, than the SPX position alone. The visualizer plots profit and loss **at expiration only** and is designed to be **set-and-forget**: once you are comfortable with the combined position, the intent is to leave it in place and let the options mature through to expiration. Holding to expiration also avoids paying the commissions you would otherwise incur to close the trade.

Key capabilities:

- Plot expiration P/L curves for complex strategies on both instruments simultaneously.
- Drag strike lines interactively on the chart to explore different positions.
- Calculate Probability of Profit (POP) and Expected Value using implied volatility.
- Load live option chains from **Tastytrade** or **Interactive Brokers**, or from a CSV file.
- Optimize strategy parameters (center, width, quantity) across thousands of combinations.
- Submit orders directly to Tastytrade or Interactive Brokers.
- Save and reload strategy configurations.

1.1 Supported Strategies

Each instrument supports the following strategy types:

Strategy	Description
None	No position on this instrument.
Call	Single call — long (debit) or short (credit).
Put	Single put — long (debit) or short (credit).
CS	Call Spread — direction selects CDS (debit) or CCS (credit).
PS	Put Spread — direction selects PDS (debit) or PCS (credit).
IC	Iron Condor — short call spread + short put spread (credit) or inverse (debit).
CCond	Call Condor — two call spreads at different strikes.
PCond	Put Condor — two put spreads at different strikes.
CButt	Call Butterfly — long/short middle call with two wing calls.
PButt	Put Butterfly — long/short middle put with two wing puts.
Straddle	Long or short straddle (ATM call + put at same strike).

Strategy	Description
Strangle	Long or short strangle (OTM call + OTM put at different strikes).
CBWB	Call Broken-Wing Butterfly — long 1 / short 2 / long 1 calls, the far wing wider.
PBWB	Put Broken-Wing Butterfly — long 1 / short 2 / long 1 puts, the far wing wider.
DBWB	Double Broken-Wing Butterfly (SPX only) — a call BWB above the center plus a put BWB below.
Custom	Up to 4 fully configurable legs (type, strike, ratio, price).

1.2 Instruments

The application always works with two instruments plus an optional hedge:

- **Inst 1 (SPX):** The primary instrument. Strike entry fields use SPX points (e.g., 5 500). BPR and P/L are in SPX dollar terms (multiplier 100).
- **Inst 2 (XSP):** The secondary instrument. Strikes are in XSP points (approx. 1/10th of SPX). A configurable *scale factor* maps XSP prices to SPX-equivalent for the combined P/L curve.
- **Additional XSP Hedge:** An optional third position on XSP using any strategy. Shown separately in the chart and included in the combined P/L and BPR totals.

2 Installation

2.1 System Requirements

Component	Minimum	Recommended
Operating System	Windows 10 (64-bit)	Windows 11 (64-bit)
Processor	Any 64-bit dual-core (~1.6 GHz)	Quad-core (faster Optimizer runs)
RAM	4 GB	8 GB
Disk space	~250 MB free	~250 MB free
Display	1600 × 900	1920 × 1080
Internet	Only for live broker data	Only for live broker data

2.2 Installing from the Installer

- Run **OptionsRiskGraphSetup.exe**.
- Accept the UAC prompt (installation goes to Program Files).
- Optionally create a desktop shortcut when prompted.
- Click *Launch SPX-XSP Options Visualizer* at the end of the wizard.

*Note: On first launch the application creates a configuration file **risk_graph_config.json** in the installation folder. Saved strategies go to the **strategies** subfolder; chain CSV files go to **chains**.*

2.3 Licensing

The application requires a valid license key. Your key is created automatically when you purchase the product on **Whop** and is shown on your Whop membership page — copy it and paste it into the activation dialog on first launch, then click *Activate*.

- **Two devices per key.** A license key may be activated on up to two computers you personally own (e.g., a desktop and a laptop). Activating a third computer is rejected until you deactivate one of the existing devices.
- **Online check each launch.** The key is re-validated when the app starts, so a canceled or expired membership stops access. A short offline grace lets the app run if your internet is temporarily unavailable.
- **Moving to a new computer.** Use *Help* → *License* → *Deactivate this device* (or reset the key from your Whop membership page), then activate on the new machine.

Note: An internet connection is required for the first activation. Your license key remains visible on your Whop membership page if you need it again.

3 Interface Overview

The main window is divided into two areas: a **left control panel** and a **chart area** on the right.

3.1 Left Control Panel (top to bottom)

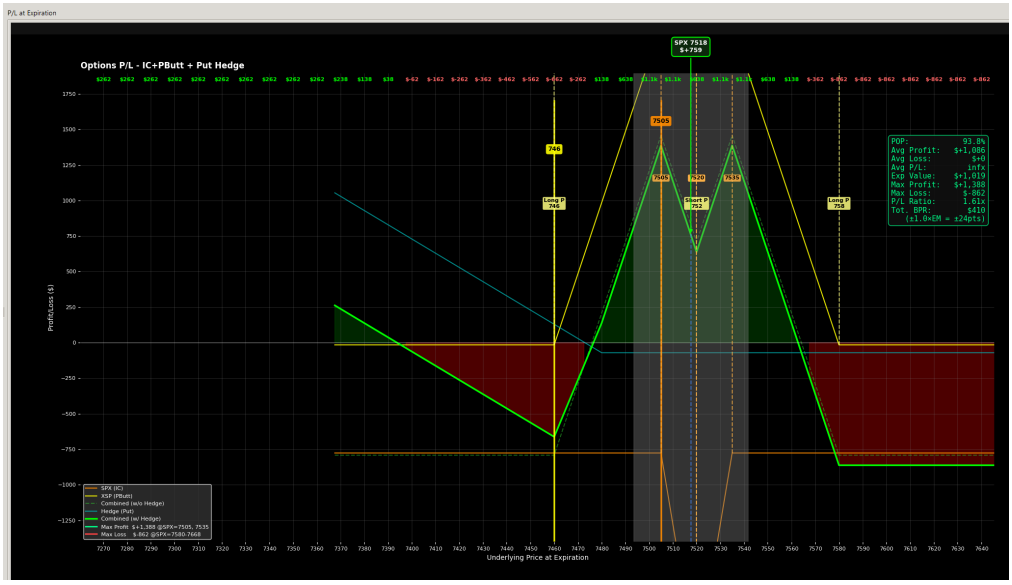
- **Data Source row** — select CSV Data or a broker account, choose expiration, load chain.
- **SPX Strategy section** — strategy type, direction, strikes, quantity, BPR.
- **XSP Strategy section** — same controls scaled to XSP.
- **Additional XSP Hedge section** — optional third position on XSP.
- **POP & IV section** — implied volatility, DTE, Expected Move multiplier.
- **Price Range section** — controls the X-axis of the chart.
- **Buttons** — Fetch Prices and Lock All (data-source row), the per-section Save buttons, and Enter Orders. Strategy load/save, the Optimizer, the Calculator, the clean-up actions, and Configuration now live on the **menu bar** (see §3.3).

3.2 Chart Area

The chart plots **P/L at expiration** (vertical axis, in dollars) against **underlying price** (horizontal axis, in SPX points). Multiple curves are drawn:

- A coloured curve for each active instrument/hedge position.
- A **green combined curve** summing all positions.
- A **grey EM band** shading the Expected Move window.
- Vertical **strike lines** with draggable flag labels.

Note: The chart auto-recalculates whenever any input field changes.



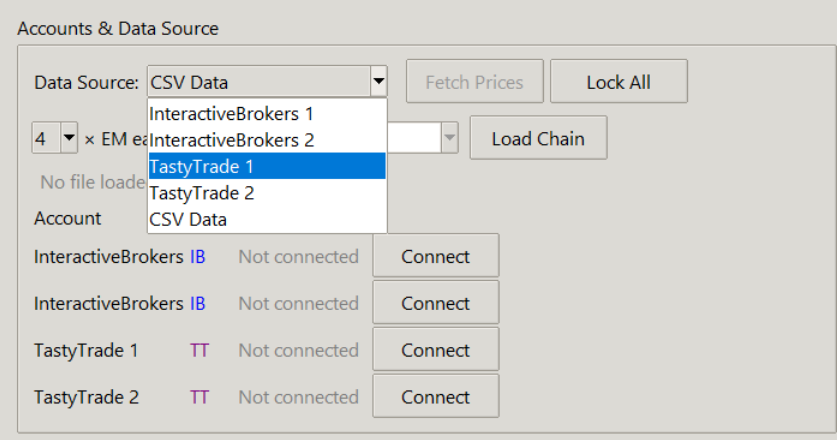
3.3 Menu Bar

The menu bar across the top of the window groups the application actions:

- **File** — Load Strategy, Save Strategy.
- **Tools** — Intrinsic Value Calculator, Optimize, Optimize DBWB.
- **Clean Up** — Clear Position, Clear All, Reset Defaults.
- **Configuration** — opens the Configuration window (see Section 13).
- **About** — Version, License, Check for Updates.

4 Data Sources

The application can pull option chain data from three sources: **CSV Data**, **Tastytrade**, and **Interactive Brokers**. Select the source from the **Data Source** dropdown at the top of the left panel.



4.1 CSV Data

Use a pre-downloaded option chain CSV file.

- Select **CSV Data** from the Data Source dropdown.
- Choose the **Expiration** date from the combo box (or type it).
- Click **Load Chain** and browse to the CSV file.
- Once loaded, the filename is shown below the button and premiums are populated automatically when strikes are entered.
- Premiums update whenever a strike changes.

Example of Bad Data from CSV File

When loading CSV data, corrupted prices can cause incorrect calculations. Bad data occurs when the calculated spread price does not match the strategy type (e.g., a debit spread shows a credit price or vice versa). This typically happens when the short leg is more expensive than the long leg. The application detects bad data and displays a yellow background and a warning triangle on the price textbox. Strategies marked with bad data are automatically skipped during optimization.



Note: The CSV must contain columns for strike, call bid/ask, and put bid/ask. The expiration date is read from the file header if present.

4.2 Data From Brokers

When connected to a Tastytrade or Interactive Brokers account, the application streams live option chain data directly from the broker. The broker account must first be configured in the **Configuration** window (see Section 13) and connected via the **Connect** button in the accounts panel.

- Select your broker account from the **Data Source** dropdown — it appears there automatically once connected.
- The **Expiration** combo box is populated with available expiry dates from the broker. Select the desired expiration.
- Prices stream automatically for all active strategy legs.
- Use **Fetch Prices** to request a fresh one-time snapshot, or let streaming keep prices updated continuously (see Section 5).

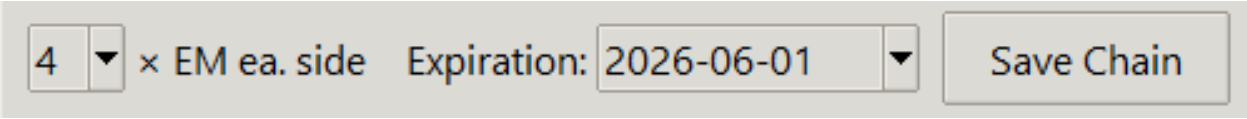
Note: To set up broker accounts, see Section 13 — Configuration.

4.3 Saving Chain Files

When connected to a broker you can save a snapshot of the current option chain to a CSV file for later offline use. To save a chain:

- Select the **Expected Move multipliers** (EM ×) you want to include in the saved chain.
- Select the desired **Expiration** date from the combo box.
- Click the **Save Chain** button.
- The file is saved to the **chains** folder inside the Data Directory and will appear automatically in the file browser next time you click **Load Chain**, making it quick to switch between saved chain files.

Broker Notes: For Interactive Brokers, it is recommended to use an Expected Move multiplier of **less than 3** to avoid fetching illiquid far-OTM strikes. In general, **TastyTrade is faster** and will return **larger option chains** compared to Interactive Brokers, making it the preferred source for comprehensive chain snapshots.



5 Fetching Prices & Streaming

Once a broker account is selected as the data source, two buttons appear in the top control row: **Fetch Prices** and **Lock All**.

5.1 Fetch Prices Button

Clicking **Fetch Prices** sends a one-time request to the broker for the current mid prices of all active strategy legs based on the strikes currently entered. The button is disabled when CSV Data is selected.

- Prices are retrieved for every active leg across SPX, XSP, and the hedge.
- After a successful fetch the price entries are updated and the chart recalculates.
- If streaming is already active, clicking the button stops it — the label changes to **Stop Streaming** while streaming is running.
- Fetch Prices also runs automatically when you first connect to a broker account.

5.2 Live Streaming

When a broker account is connected the application can stream live bid/ask prices continuously, updating all leg premiums in real time.

- Click **Fetch Prices** to start streaming — the button label changes to **Stop Streaming**.
- Prices update automatically as the market moves; the chart recalculates each time.
- Click **Stop Streaming** to halt and return to static prices.
- Streaming restarts automatically if strikes or the expiration date are changed.



Example of Fresh Price



Example of Stale Price

5.3 Price Locks

Each strategy price field has a **lock icon** button next to it. Locks let you manually fix a premium and prevent it from being overwritten by the next fetch or streaming update.

- **Unlocked** (open padlock) — the price is fetched automatically from the broker or CSV. The entry field is disabled (grey).
- **Locked** (closed padlock, white background) — the price is fixed at the value you typed. Fetch Prices and streaming will not overwrite it.
- Useful when you want to model a fill price that differs from the current mid, or to hold one leg constant while adjusting others.

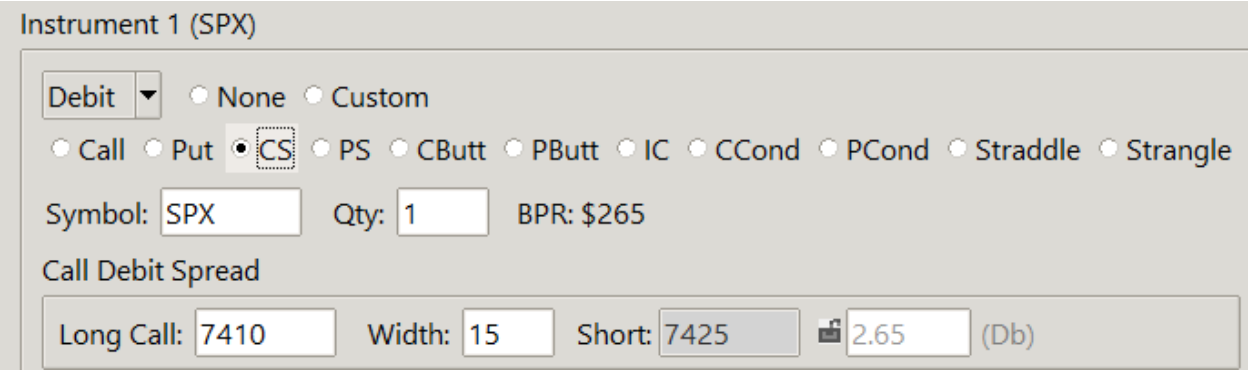
The **Lock All** button toggles every active price lock at once — convenient for freezing all prices before manually editing, or releasing them all to resume live updates. The label switches between **Lock All** and **Unlock All** depending on the current state.

6 SPX Strategy Setup

The **SPX section** controls the primary (Inst 1) position. All strikes are entered in SPX points; the multiplier is 100.

6.1 Selecting a Strategy

- Click the radio button for the desired strategy (None, Call, Put, CS, PS, IC, etc.).
- **Direction** — choose *Credit* (sell premium) or *Debit* (buy premium). Direction determines whether spreads and condors are credit or debit structures.
- **Qty** — number of spreads/contracts. Multiplies the entire position.



6.2 Strike Entry

Each strategy reveals the relevant strike fields. Enter strikes as whole numbers on the SPX grid (multiples of 5).

Strategy	Strike Fields	Notes
Call / Put	Strike	Single-leg at this strike.
CS / PS	Anchor strike + Width	Width = distance between the two legs.
IC	Call anchor + Put anchor + Width	Both spreads use the same width.
CCond / PCond	Two anchor strikes + Width	Inner and outer spread anchors.
CButt / PButt	Middle strike + Wing width	Wings are middle ± width.
Straddle	Strike	Both call and put at this strike.
Strangle	Put strike + Call strike	Two separate strikes.
CBWB / PBWB	Near + W + Broken	Long 1 at Near, short 2 at W further out (calls up, puts down), long 1 at Broken beyond that. Price is the net (+ credit / – debit).

Strategy	Strike Fields	Notes
DBWB	Center + Offset + W + Broken	SPX only. Call and put BWBs whose near legs sit at Center ± Offset.
Custom	Up to 4 legs	See Section 6.3.

Note: If a CSV chain is loaded, premiums fill in automatically when you tab out of a strike field. Premiums can also be typed manually.

6.3 Custom Strategy (4 Legs)

Select **Custom** to configure up to four independent legs:

- **L / S** — toggle Long or Short for this leg.
- **Type** — C (call) or P (put).
- **Strike** — the strike price.
- **× (Ratio)** — leg multiplier relative to the strategy quantity.
- **Price** — premium. Click the lock icon to override the auto-filled price.

Note: Custom strategies are excluded from the optimizer.

6.4 BPR Display

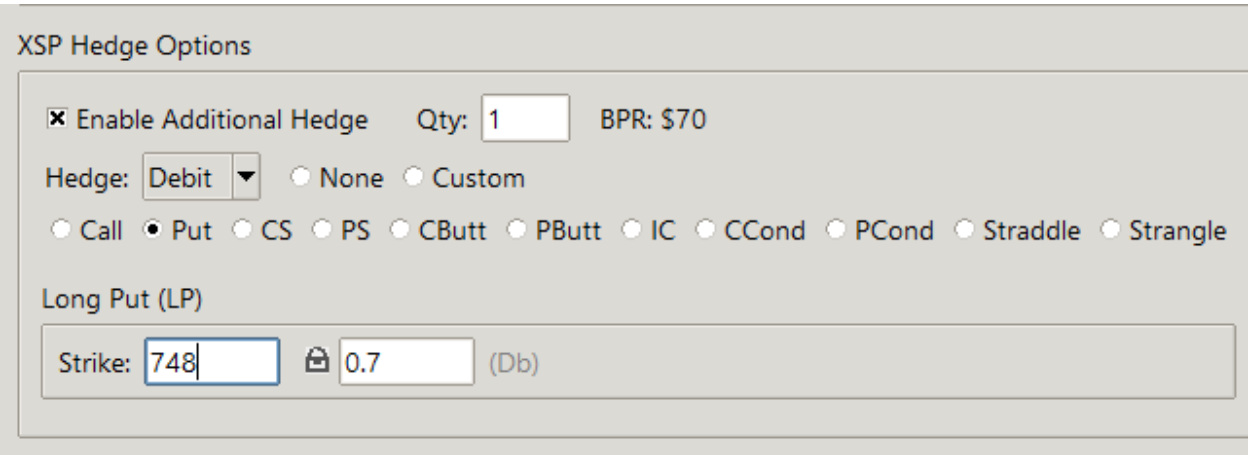
The **BPR** (Buying Power Reduction) label below the strategy row shows the margin requirement for the current configuration. The **Tot. BPR** at the bottom of the panel sums BPR across all instruments including the hedge.

7 XSP Strategy Setup

The **XSP section** controls the secondary (Inst 2) position. It has identical controls to the SPX section. Strikes are in XSP points (approximately 1/10th of SPX). The **Scale Factor** field converts XSP prices to SPX-equivalent for the combined P/L chart.

- Default scale factor: **10** (XSP × 10 ≈ SPX).
- Adjust if using a non-standard ratio.
- The XSP P/L curve is plotted after scaling, so it overlaps meaningfully with the SPX curve.

Note: The combined (green) curve sums the SPX and XSP P/L values at each price point, allowing you to see the net position across both instruments.

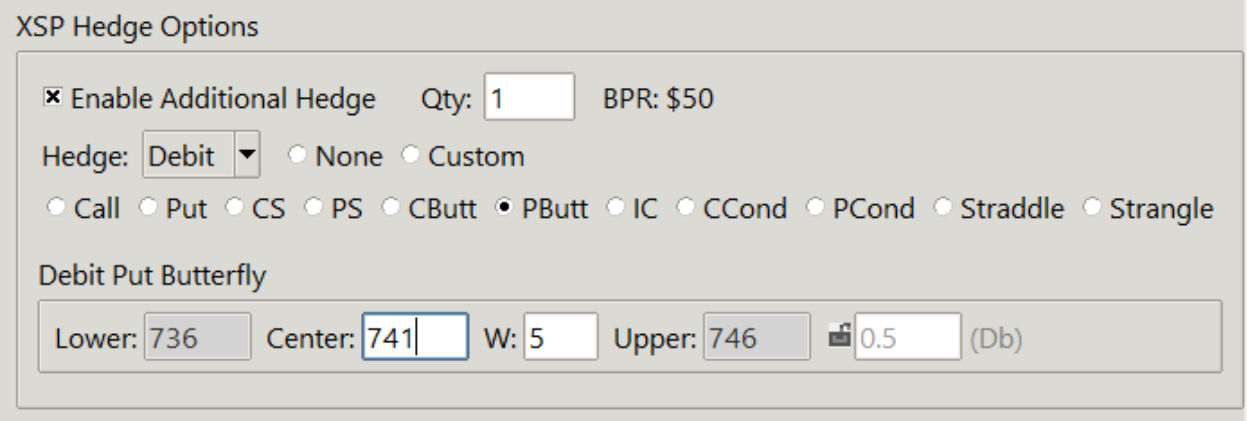


8 Additional XSP Hedge

The **Additional XSP Hedge** section adds a third, independent position on the XSP instrument. It is useful for hedging an existing SPX/XSP position with a separate structure (e.g., a long put spread or long call).

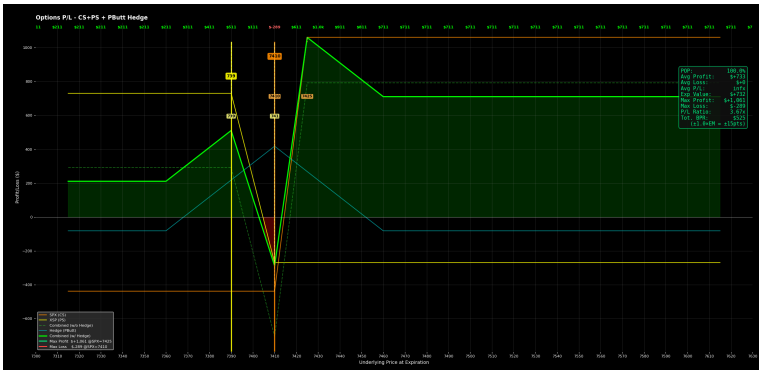
8.1 Enabling the Hedge

- Check **Enable Additional Hedge** to activate the section.
- Select strategy, direction, and enter strikes exactly as in the XSP section.
- Enter a separate **Qty** for the hedge.
- The hedge BPR is shown inline and included in **Tot. BPR**.



8.2 Hedge in the Chart

When enabled, the hedge P/L appears as a separate curve and is also included in the combined green curve. This lets you immediately see how the hedge affects the overall risk profile. The hedge curve is displayed in **teal** color.



Note: The hedge uses the XSP scale factor for conversion to SPX-equivalent in the combined curve.

9 Chart Interaction

9.1 Reading the Chart

The P/L chart shows each position's profit or loss at expiration for every underlying price on the X-axis. Positive values are profits; negative values are losses.

- The **grey shaded band** represents the Expected Move window (S0 ± EM × IV_move). Most outcomes are expected to fall in this region.
- The **vertical dashed lines** mark each active strike.
- The **flag labels** on the lines show the strategy name and strike value.

9.2 Dragging Strikes

Strike lines are interactive — drag them with the mouse to explore different positions:

- **Solid line with large flag** (leftmost strike) — drags the *entire combo*. All strikes shift together, preserving the spread width.
- **Dashed lines** — drag only that individual leg. The other leg stays fixed, changing the spread width.

Note: Premiums update automatically after dragging if a CSV chain is loaded.

9.3 POP and Statistics Box

Below the chart (or in a stats panel) the following metrics are displayed:

- **POP** — Probability of Profit, calculated from the log-normal price distribution.
- **Max Profit / Max Loss** — within the price range.
- **Exp Value** — probability-weighted average P/L at expiration.
- **Avg Profit / Avg Loss** — average P/L in winning and losing scenarios.
- **Avg P/L Ratio** — average profit divided by average loss (absolute).

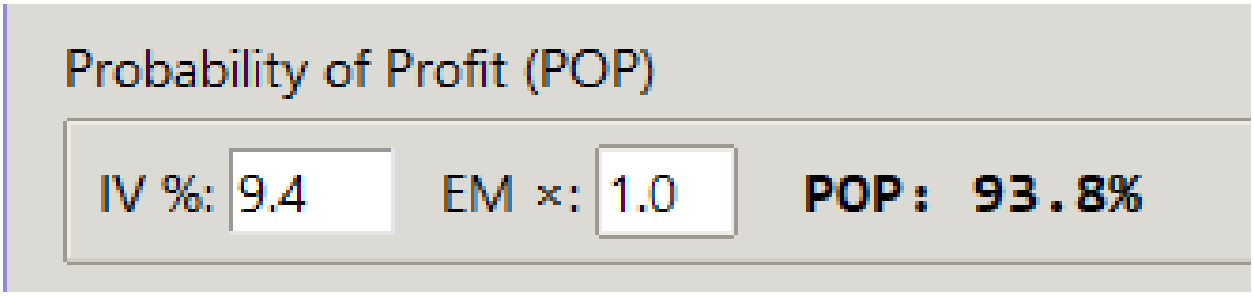
10 POP & IV Settings

These fields control the probability and Expected Move calculations.

Field	Description
IV %	Implied volatility as a percentage (e.g., 16 for 16% IV). Used to compute the Expected Move and POP. Loaded automatically from the broker chain when streaming data is active.
DTE	Days to expiration. **Calculated automatically by the application from the Expiration date field.** Together with IV%, determines the size of the Expected Move cone.
EM ×	Expected Move multiplier. Scales the EM band on the chart. 1.0 = one standard-deviation move; 2.0 = two standard deviations.
POP	Displayed read-only. Probability that the combined position expires with P/L ≥ 0 under the log-normal model.

The Expected Move formula used is:

```
EM = Underlying Price × (IV% / 100) × √(DTE / 365)
```



11 Strategy Optimizer

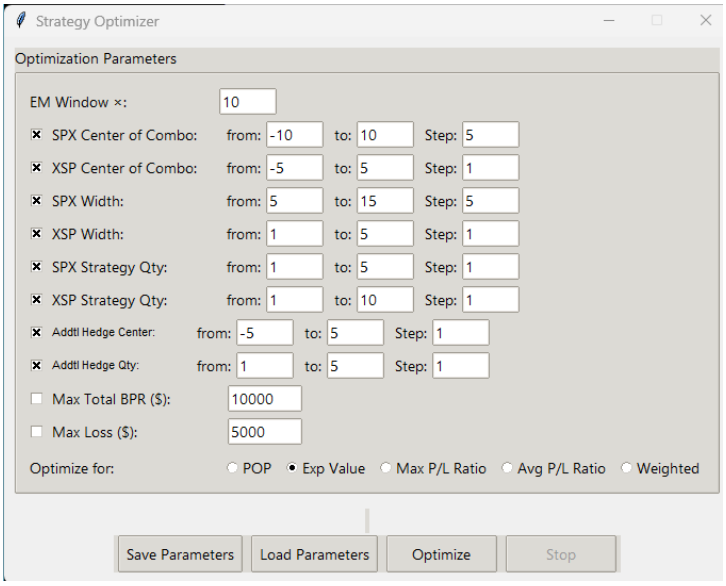
The Optimizer searches combinations of center offsets, spread widths, and quantities to find the configuration that best satisfies a chosen metric. Open it from **Tools** → **Optimize** on the menu bar.

11.1 Optimizer Parameters Window

The window is laid out top-to-bottom as follows:

- **EM Window ×** — multiplier used both for the chart EM band and as an optional strike-bounding filter (see §11.4).
- **SPX / XSP Center of Combo** — range of offsets (in points) applied to the anchor strikes. A value of +5 shifts all SPX strikes 5 points higher.
- **SPX / XSP Width** — spread-width range (not shown for None, Custom, Call, Put, Straddle, Strangle or the broken-wing butterflies; Custom, CBWB, PBWB and DBWB positions are held fixed — to optimize a DBWB use **Tools** → **Optimize DBWB**, which sweeps its center, offset, W, broken wing and quantity).
- **SPX / XSP Strategy Qty** — number of spreads to try.
- **Additional Hedge rows** — visible only when the hedge is enabled; same centre/width/qty parameters for the hedge.
- **Max Total BPR ($)** — if checked, skip combinations whose total BPR exceeds this dollar amount.
- **Max Loss ($)** — if checked, skip combinations whose maximum loss exceeds this dollar amount.
- **Optimize for** — the metric to maximise (see §11.3).
- **Weighted** — when selected, reveals weight sliders for a custom composite score.

Note: Each row has a checkbox to enable/disable that parameter. Disabled parameters use the current value (no range is swept).



11.2 Running the Optimizer

- Set the desired parameter ranges.
- Click **Optimize**. The button disables and a progress counter appears.
- Click **Stop** at any time to halt early — partial results are still applied.
- When complete, the best result is applied to the main window and a summary dialog appears.
- A CSV file of all tested combinations is saved to the **Optimizer** folder.

11.3 Optimization Targets

Target	Description
POP	Maximise Probability of Profit.
Exp Value	Maximise probability-weighted average P/L (POP × avg profit + (1–POP) × avg loss).
Max P/L Ratio	Maximise (max profit) ÷ |max loss|.
Avg P/L Ratio	Maximise (avg profit) ÷ |avg loss|.
Weighted	Composite score: w■·POP + w■·Exp Value + w■·Max P/L + w■·Avg P/L. Weights must sum to 1.0; w■ is auto-calculated.

11.4 EM Bounds Filtering

When a valid underlying price and IV% are set, the optimizer can enforce that all strikes fall within the Expected Move window. This prevents the optimizer from placing strikes far outside the likely price range. Combinations with strikes outside **S0 ± EM_Window × EM** are skipped automatically.

11.5 Saving and Loading Parameters

- **Save Parameters** — saves the current optimizer settings to a JSON file for reuse.
- **Load Parameters** — restores a previously saved set of settings.
- The last-used parameters are saved automatically and restored next time the optimizer opens.

11.6 Optimizer Results Files

Every optimizer run writes a timestamped CSV file to the **Optimizer** folder inside the Data Directory, containing all tested combinations and their metrics. These files can be opened in Excel for further analysis.

12 Saving & Loading Strategies

12.1 Auto-Save

The application saves the full configuration to **risk_graph_config.json** automatically on exit and reloads it on the next launch. This includes all strategy parameters, strikes, quantities, IV%, DTE, and display settings.

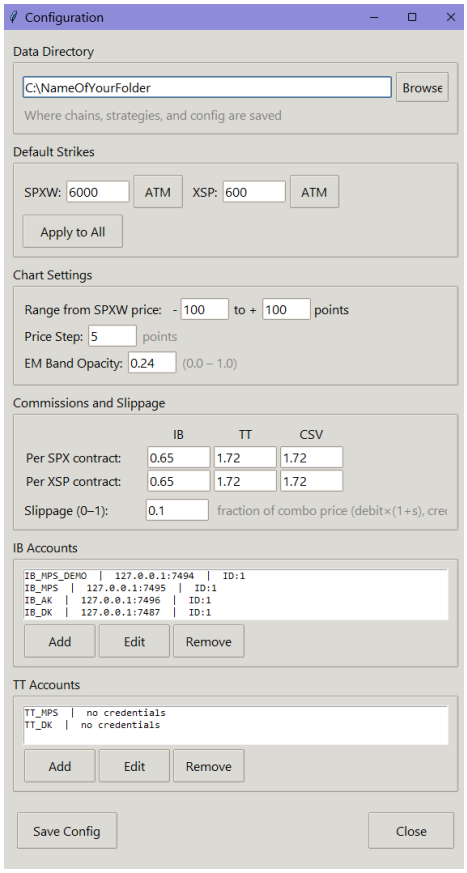
12.2 Named Strategies

Individual strategy configurations can be saved and recalled:

- Use **File** → **Save Strategy** on the menu bar, or the **Save** button within a strategy section.
- Enter a name — the file is written to the **strategies** folder.
- Use **File** → **Load Strategy** and select a file to restore a saved configuration.

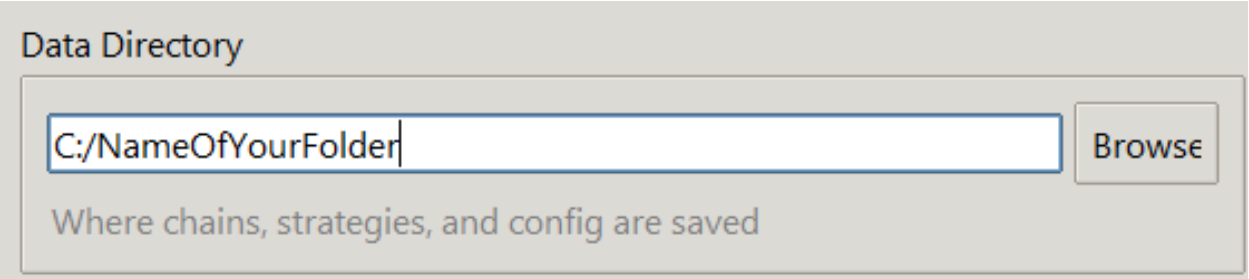
13 Configuration

Open the Configuration window from the **Configuration** menu on the menu bar. All settings are persisted to the config file when you click **Save Config**.



13.1 Data Directory

Sets the folder where chains, strategies, optimizer results, and the config file are saved. Use the **Browse…** button to choose a different location. The path is stored separately so the application can find its data on next launch.



13.2 Default Strikes

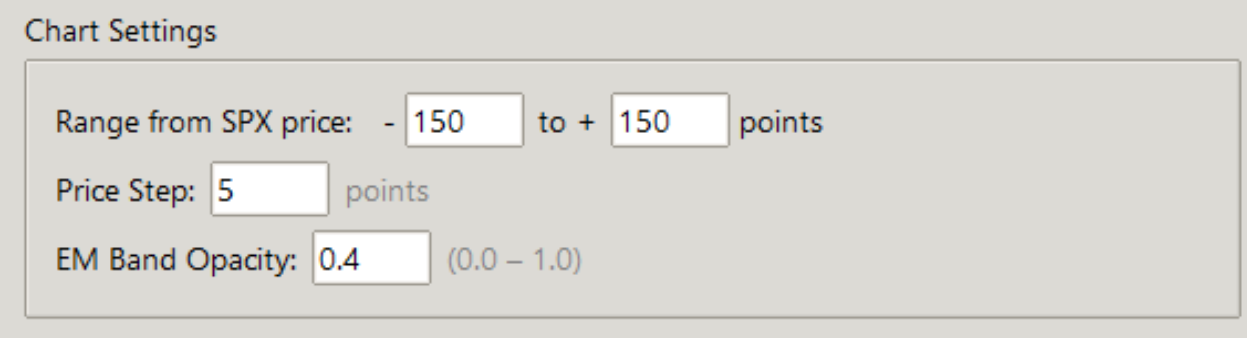
The default anchor strike applied to all legs when the strategy is first selected or when **Apply to All** is clicked.

- Enter a fixed value for SPX and XSP, or click **ATM** to set the current at-the-money strike automatically (requires an active price feed).
- **Apply to All** — pushes the default strikes into every leg of both instruments immediately.

13.3 Chart Settings

Controls the X-axis span and appearance of the P/L chart.

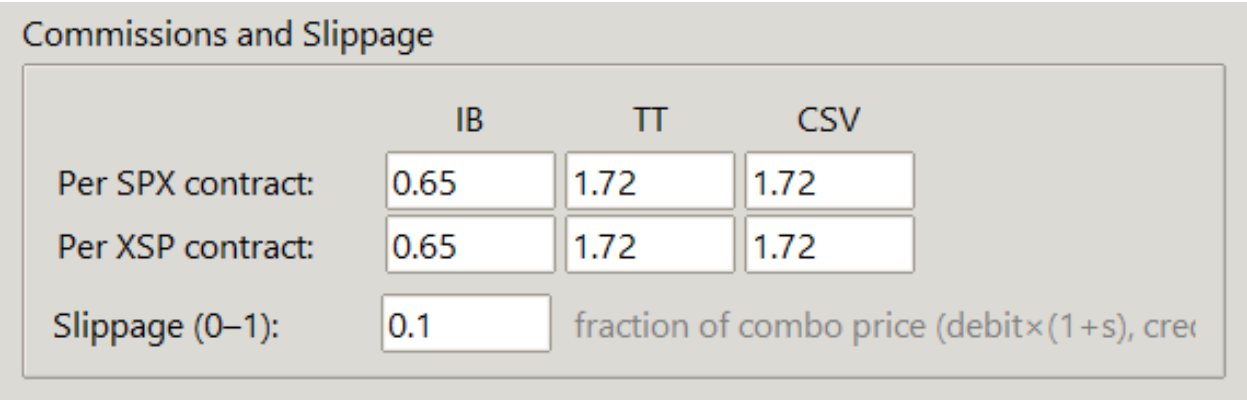
Field	Description
Range from SPX price	Sets how far left (–) and right (+) of the current underlying price the chart extends, in points.
Price Step	Spacing between P/L calculation points. Smaller step = smoother curve, slower calculation.
EM Band Opacity	Controls how prominently the grey Expected Move shading appears on the chart (0.0 = invisible, 1.0 = fully opaque).



13.4 Commissions and Slippage

Commission rates are set per data source (IB, TT, CSV) and per instrument. They are subtracted from P/L curves and included in BPR calculations. Total commission for a position is the per-contract rate multiplied by the number of legs and the quantity.

- **Per SPX contract** — fee charged per SPX (Instrument 1) option contract.
- **Per XSP contract** — fee charged per XSP (Instrument 2) option contract; this rate also applies to the Additional XSP Hedge.
- **Slippage (0–1)** — fraction of the combo price added as a fill-cost penalty (debit positions cost more, credit positions receive less). E.g., 0.05 = 5% slippage.



13.5 Adding Broker Accounts

Initially you will need to add at least one broker account if you want to get data from a broker instead of entering prices manually.

13.5.1 IB Accounts

Prerequisite: TWS (Trader Workstation) or IB Gateway must be running and logged in before the application can connect.

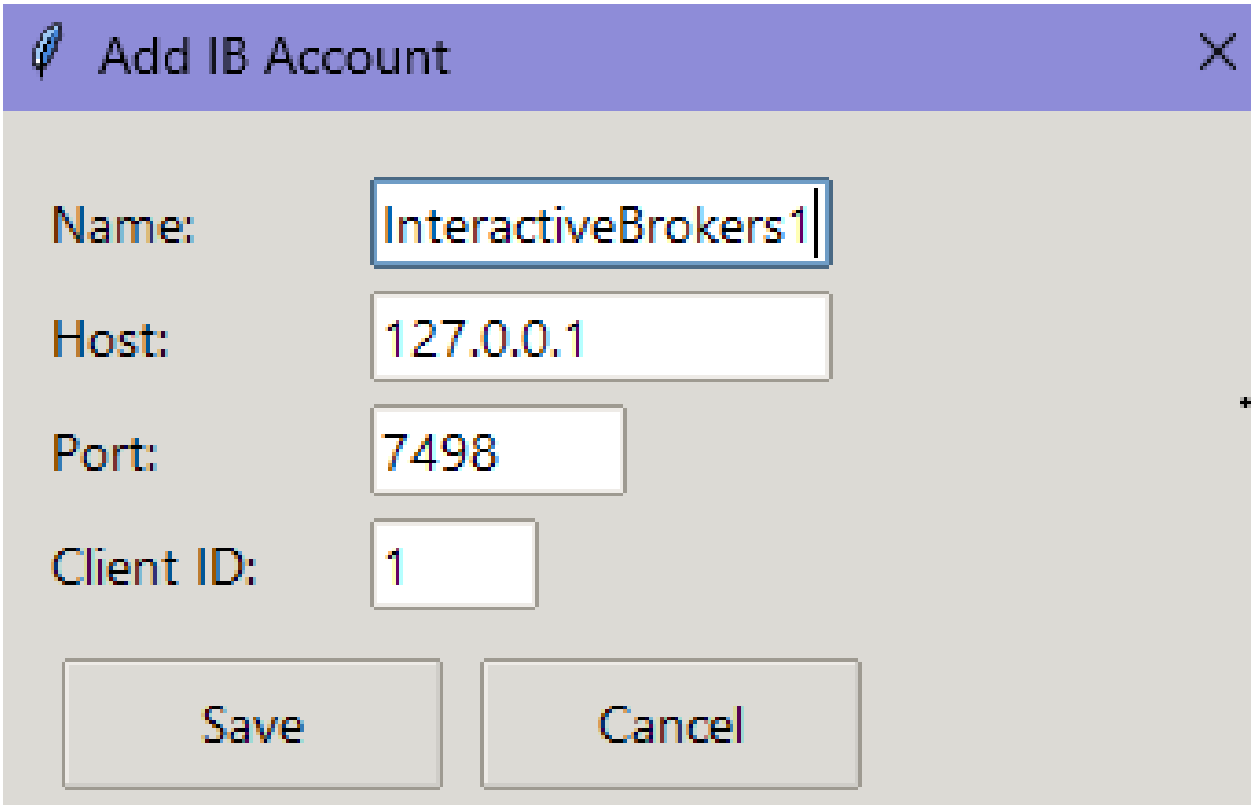
Before adding an account, open TWS and go to **Edit** → **Global Configuration** → **API** → **Settings**. Ensure **Enable ActiveX and Socket Clients** is checked and note the **Socket port** number — you will need it when filling in the account details below.

Click **Add** to register a new IB account. Fill in:

- **Name** — a label for this account (e.g. IB_Main).
- **Host** — IP address where TWS/Gateway is running (default: 127.0.0.1).
- **Port** — the API port noted from TWS API Settings above.
- **Client ID** — a unique integer identifying this connection (default: 1).

• Click **OK** to save. The account name then appears in the **Data Source** dropdown.

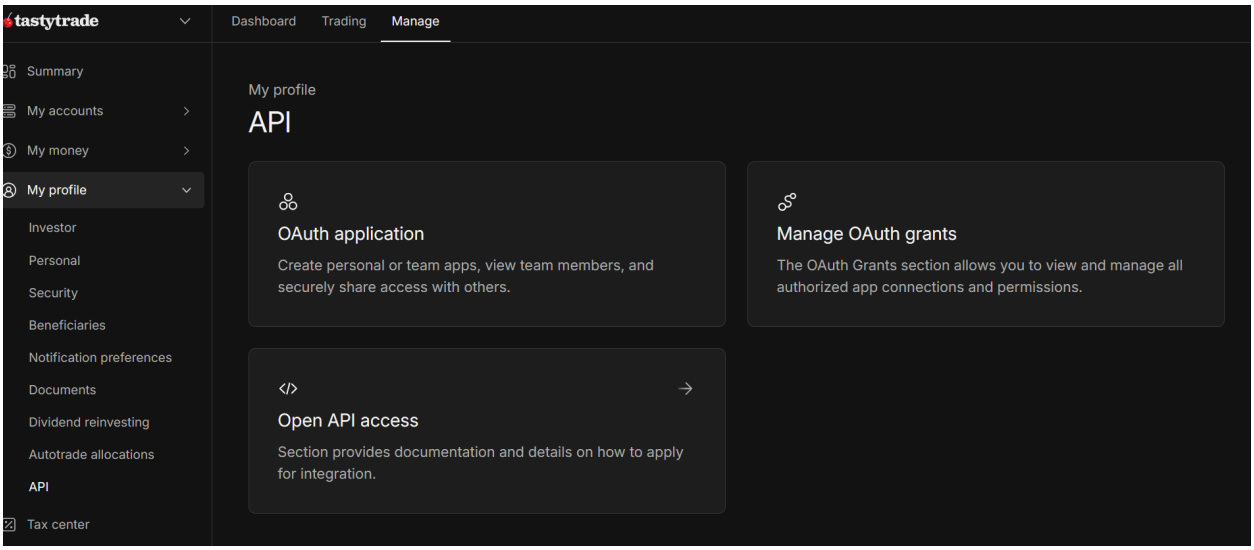
• Use **Edit** or **Remove** to modify or delete an existing account.



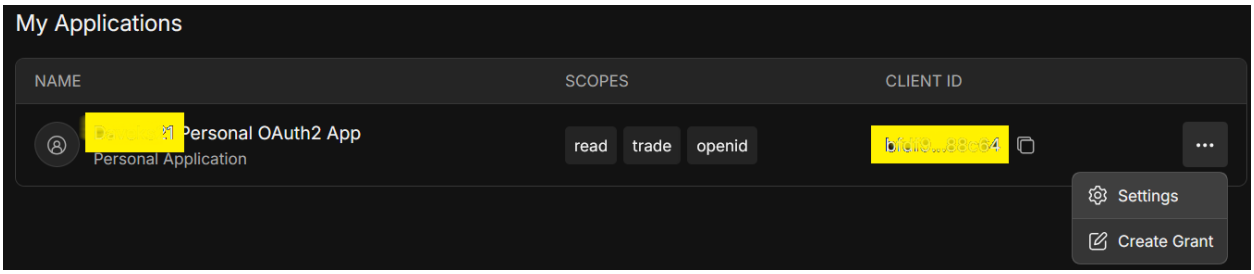
13.5.2 TT Accounts

Prerequisite: Before registering a Tastytrade account in this application, you must create an OAuth client in your Tastytrade profile to obtain the **Client Secret** and **Refresh Token**.

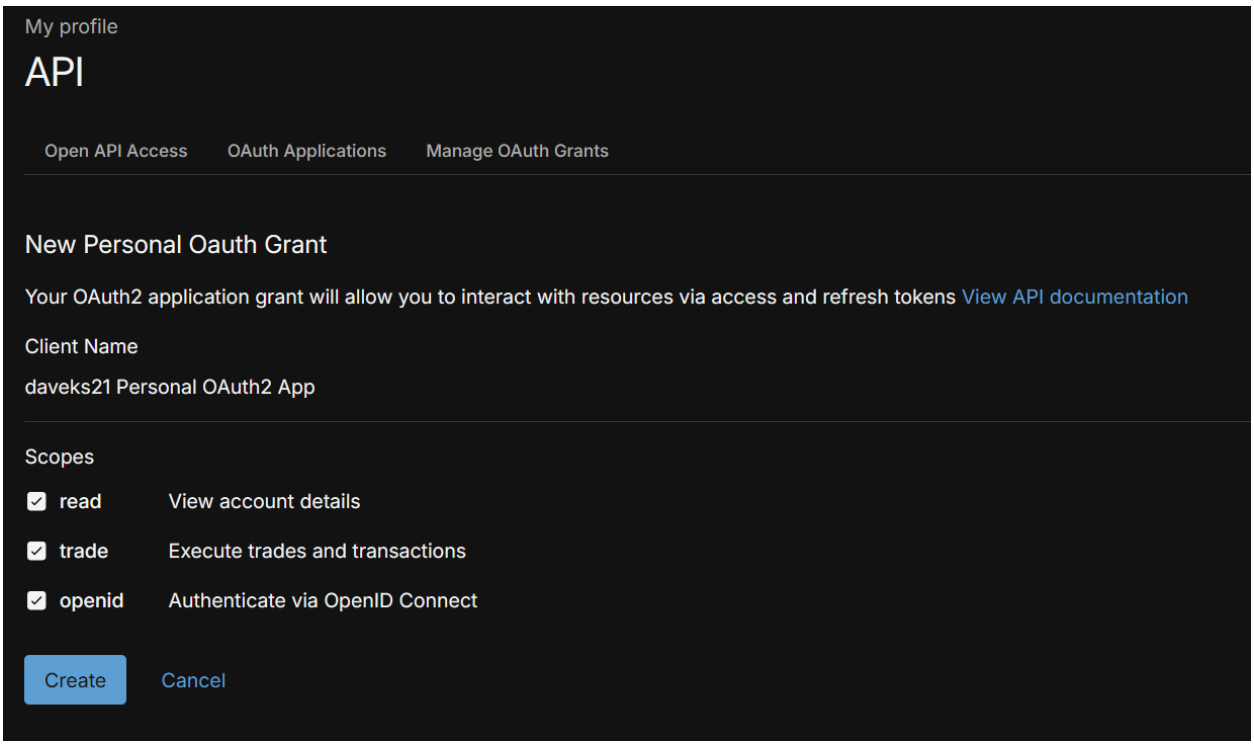
Go to the **TastyTrade website** and log in. In your **Profile**, navigate to the **API section**.



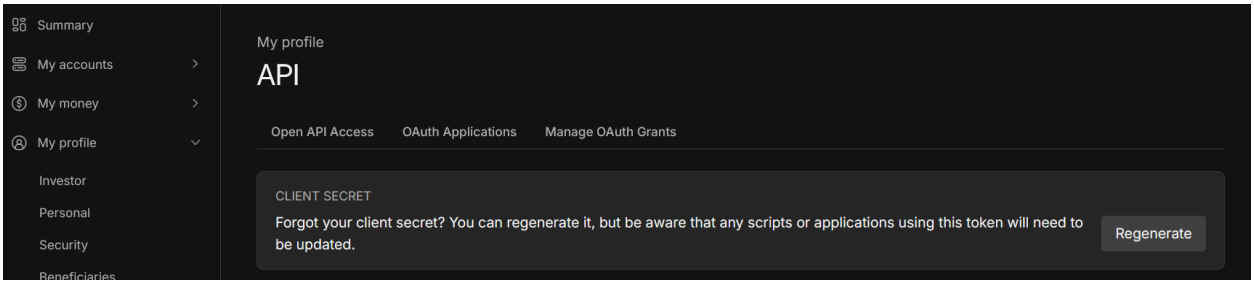
In the API section, navigate to **OAuth Applications** and click **+New OAuth Client** to create a new application.



After creating the OAuth client, go to **Manage Grants** and select the required scopes: **read**, **trade**, and **openid**. These permissions allow the application to access your account data and submit orders.

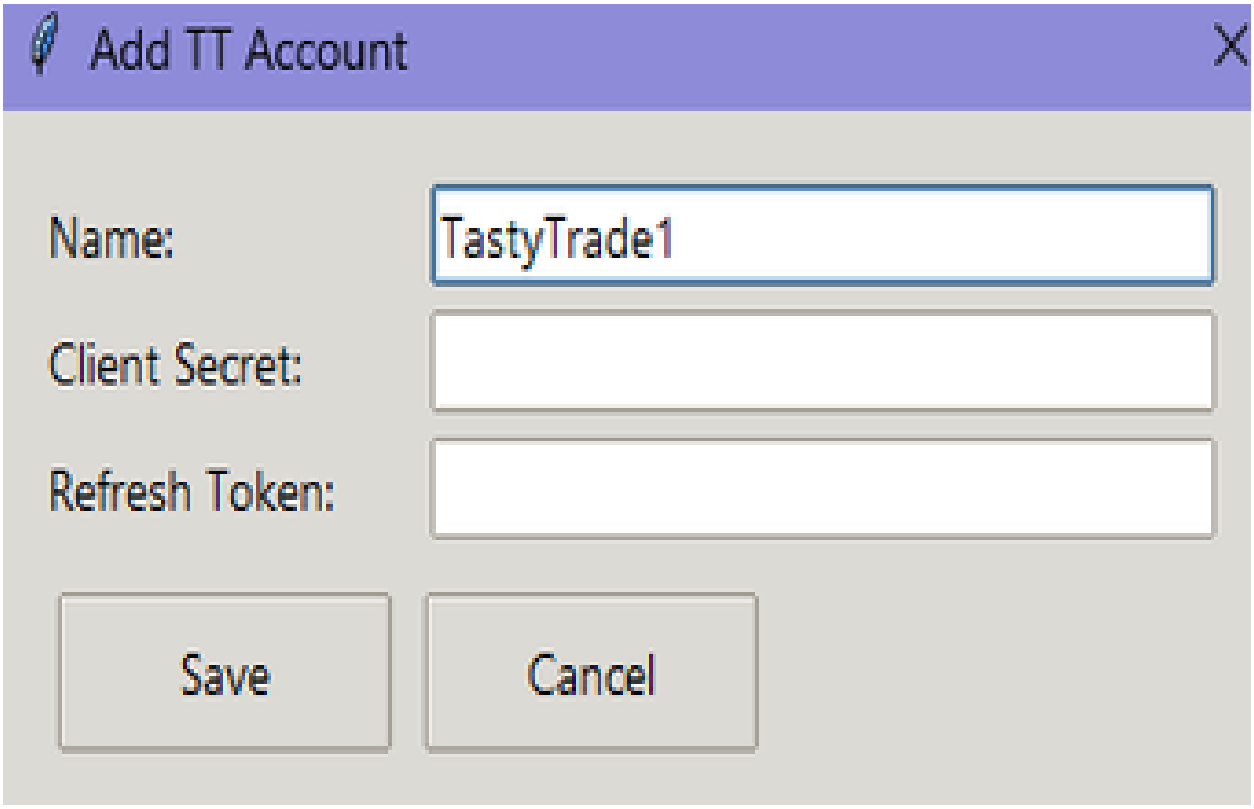


From your OAuth client settings, copy the **Client Secret** and **Refresh Token**. These values are required to register the account in this application.



Back in this application, click **Add** in the TT Accounts section to register a new Tastytrade account. Fill in:

– **Name** — a label for this account (e.g. TT_Main).
– **Client Secret** — the client secret from your TastyTrade OAuth client.
– **Refresh Token** — the refresh token from your TastyTrade OAuth client.



Click **OK** to save the account. The account name then appears in the **Data Source** dropdown. Once selected and connected, the option chain will stream automatically from Tastytrade.

13.6 Save Config

Click **Save Config** to write all current configuration values (data directory, default strikes, chart settings, commissions, slippage, instrument alignment, and all broker accounts) to the config file in the Data Directory. Settings are also saved automatically when the application exits normally. Closing the Configuration window triggers an immediate chart recalculation to reflect any changes.

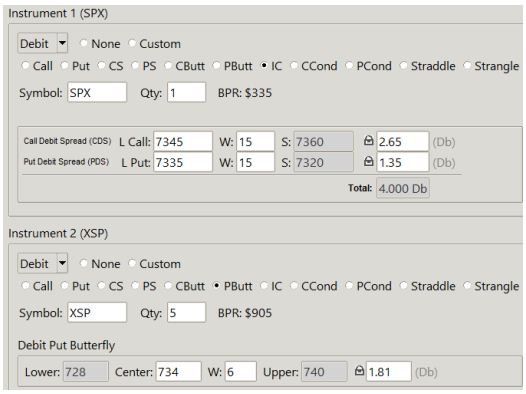
14 Examples

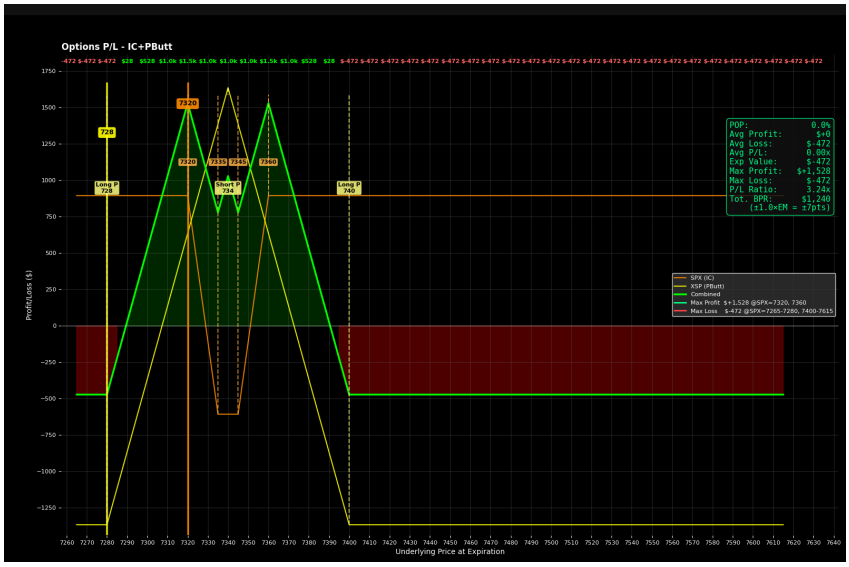
Here are some examples of possible strategies.

14.1 Mountain Range

The Mountain Range combines a **debit Iron Condor on SPX** with a **debit Put Butterfly on XSP**, creating a wide profit zone with a distinctive multi-peak profile on the risk graph.

- **SPX Debit Iron Condor** — width 15. Profits when the underlying stays within the inner strikes at expiration.
- **XSP Debit Put Butterfly** — width 6, quantity 5. Adds a central profit peak that enhances the combined P/L near the current price.
- Quantities and widths can be adjusted to match the user's risk preference.

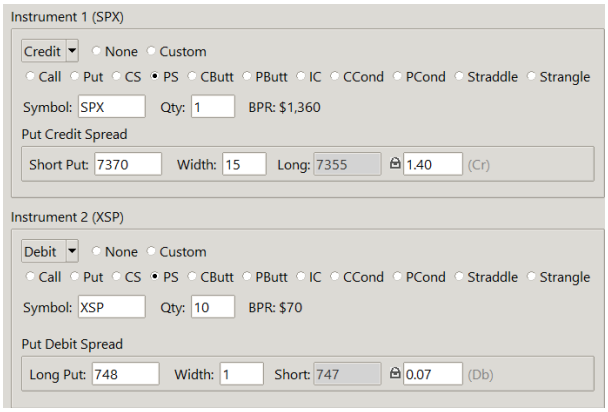




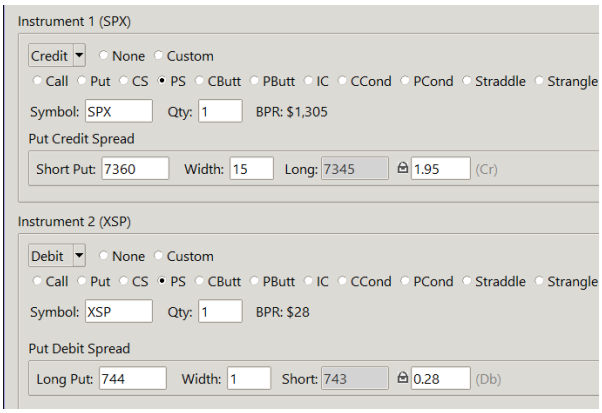
14.2 Hedged Put Spread

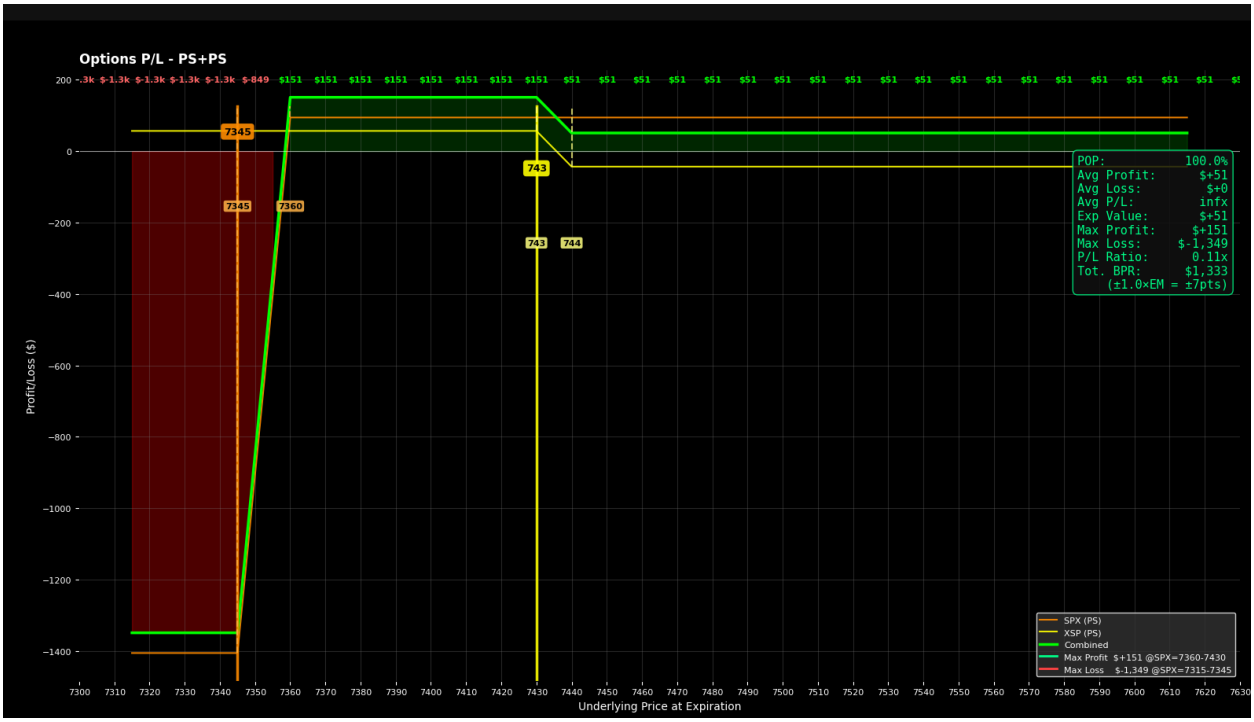
The Hedged Put Spread pairs a **put credit spread on SPX** with a **hedging position on XSP**, reducing the downside risk of the spread while preserving most of the credit received.

- The SPX put credit spread collects premium and profits if the market stays above the short strike at expiration.
- The XSP hedge offsets losses on the downside, capping the maximum loss significantly compared to an unhedged spread.
- Quantities and strikes can be adjusted to control the cost of the hedge versus the degree of downside protection desired.





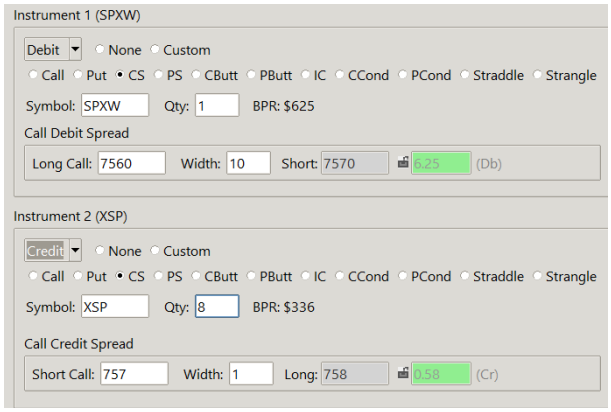

Instrument 1 (SPX)
Credit
None
Custom
Call
Put
CS
PS
CButt
PButt
IC
CCond
PCond
Straddle
Strangle
Symbol: SPX
Qty: 1
BPR: $1,305
Put Credit Spread
Short Put: 7360
Width: 15
Long: 7345
1.95
(Cr)
Instrument 2 (XSP)
Debit
None
Custom
Call
Put
CS
PS
CButt
PButt
IC
CCond
PCond
Straddle
Strangle
Symbol: XSP
Qty: 1
BPR: $28
Put Debit Spread
Long Put: 744
Width: 1
Short: 743
0.28
(Db)


Options P/L - PS+PS
Profit/Loss ($)
Underlying Price at Expiration
POP: 100.0%
Avg Profit: $+51
Avg Loss: $+0
Avg P/L: infx
Exp Value: $+51
Max Profit: $+151
Max Loss: $-1,349
P/L Ratio: 0.11x
Tot. BPR: $1,333
(±1.0×EM = ±7pts)
SPX (PS)
XSP (PS)
Combined
Max Profit $+151 @SPX=7360-7430
Max Loss $-1,349 @SPX=7315-7345

14.3 Mutant Butterfly

The Mutant Butterfly is a variation on the Broken Wing Butterfly. Unlike a standard symmetric butterfly, the wings are intentionally unequal, creating an asymmetric profit zone that shifts the risk/reward profile in a directional or neutral bias favored by the trader.

- Adjusting the wing widths moves the peak profit zone up or down relative to the center strike.
- One side of the butterfly carries more risk than the other, but the trade-off is a wider profit zone or reduced cost on the favored side.
- Quantities and wing widths can be tuned to control maximum loss and the shape of the profit curve.





15 Quick Reference

15.1 Typical Workflow

Step	Action	Details
1	**Select data source**	Choose CSV Data or a broker account.
2	**Load chain / set expiration**	Click Load Chain (CSV) or pick an expiry from the dropdown (broker).
3	**Set IV% and DTE**	Enter current implied volatility and days to expiration.
4	**Choose SPX strategy**	Select strategy type, direction, enter strikes, set quantity.
5	**Choose XSP strategy**	Same as SPX; adjust scale factor if needed.
6	**Enable hedge (optional)**	Check Enable Additional Hedge and configure.
7	**Review chart**	Check the combined P/L curve, EM band, POP, and BPR.
8	**Optimise (optional)**	Open Tools → Optimize, set ranges, click Optimize.
9	**Submit order (optional)**	Click Orders and submit via broker.

15.2 Optimizer Quick-Start

- Open Tools → Optimize → set EM Window × to 10–15.
- Enable SPX/XSP Center of Combo (e.g., –10 to +10, step 5).
- Enable Width if applicable (e.g., 5 to 20, step 5).
- Enable Qty (e.g., 1 to 5, step 1).
- Optionally enable Max Loss filter.
- Select **Exp Value** as the target.
- Click Optimize and wait.

15.3 Drag-Line Behaviour Summary

Line Style	Location	Effect
Solid line + large flag	Leftmost strike	Moves entire combo (all strikes shift together)
Dashed line	Any other strike	Moves only that leg (changes spread width)

15.4 Common Issues

Symptom	Resolution
Optimizer finds no results	All combinations were filtered out. Widen the EM Window ×, relax the Max Loss / BPR limits, or expand the center/width ranges.
Premiums show 0 or empty	No chain is loaded, or the strike is off the chain grid. Load a chain CSV or connect to a broker.
BPR shows 0	The strategy has no short legs (e.g., pure debit) or commission rates are not configured.
Chart not updating	Click Calculate, or check that all required strike fields are filled.
Broker connection fails	For IB: verify TWS/Gateway is running and the port matches. For Tastytrade: check credentials in Settings.

Appendix A File & Folder Reference

Path (relative to install folder)	Purpose
risk_graph_config.json	Main config — auto-saved on exit, restored on launch.
strategies*.json	Saved strategy configurations.
chains*.csv	Option chain CSV files.
Optimizer\optimize_*.csv	Optimizer result files (one per run).
Optimizer\Parameters*.json	Saved optimizer parameter sets.
Optimizer\Parameters_last_used.json	Auto-saved last optimizer parameters.

Appendix B Strategy Code Reference

Code	Full Name	Notes
CS	Call Spread	CDS (debit) or CCS (credit)
PS	Put Spread	PDS (debit) or PCS (credit)
IC	Iron Condor	CCS + PCS (credit) or CDS + PDS (debit)
CCond	Call Condor	Two call spreads
PCond	Put Condor	Two put spreads
CButt	Call Butterfly	LCButt (debit) or SCButt (credit)
PButt	Put Butterfly	LPButt (debit) or SPButt (credit)
CBWB	Call Broken-Wing Butterfly	Long 1 / short 2 / long 1 calls
PBWB	Put Broken-Wing Butterfly	Long 1 / short 2 / long 1 puts
DBWB	Double Broken-Wing Butterfly	Call BWB + put BWB (SPX only)
CCS	Credit Call Spread	Short call at anchor, long call above
PCS	Credit Put Spread	Short put at anchor, long put below
CDS	Debit Call Spread	Long call at anchor, short call above
PDS	Debit Put Spread	Long put at anchor, short put below